

PRINCE SULTAN UNIVERSITY · COMPUTER SCIENCE DEPARTMENT

Social Implications of Artificial Intelligence & Expert Systems

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8-minute presentation

0:30

Topic Introduction

Every time an algorithm grants a loan, rejects a job application, or triages a patient, it makes a decision once made by people, without being able to explain its reasoning. AI does the work of computation and reasoning; Expert Systems imitate human expertise in a given field. As both take on more autonomy, their impact spreads well beyond any single industry.

Adoption brings real gains, but also privacy, discrimination, accountability, employment, cybersecurity, and sustainability concerns. This paper investigates the ethical, societal, and sustainability impact of AI and Expert Systems, and how they can be developed and adopted responsibly.

THREE RESEARCH QUESTIONS

- 1 What are the main social implications of AI and Expert Systems?
- 2 What ethical problems may arise because of their extensive adoption?
- 3 How can these technologies be responsibly developed and adopted?

1:00

Methodology

This study takes a literature review (secondary research) approach, synthesizing existing studies and reports on the societal impact of AI and Expert Systems from credible international organizations and peer-reviewed journals.



Secondary research

A literature review synthesizing existing studies and reports rather than collecting new primary data.



Keyword search

AI Ethics, Social Implications, Sustainability, Bias, Privacy, and AI Governance, filtered for credibility and relevance.



Three ethical lenses

Utilitarianism, Kantian Ethics, and Social Contract Theory frame the evaluation of every finding.

Sources included WHO, UNESCO, NIST, Stanford HAI, and recent peer-reviewed studies by Wang et al. (2024) and Jain et al. (2023), found via keywords such as AI Ethics, Social Implications, Sustainability, Bias, Privacy, and AI Governance, filtered for credibility and relevance. Findings were then evaluated through the three ethical lenses above.

Findings & Analysis

WORKFORCE SENTIMENT · STANFORD HAI (2024)

Expect AI to affect their
job in 5 years



Fear their job could be
replaced by AI



Ethical Implications **ETHICS**

AI ethics comes down to fairness, accountability, and privacy. WHO (2021) flags low-quality data, biased clinical data, and improper collection as recurring problems: biased inputs don't just persist, they worsen existing disparities. A documented case: a health care algorithm used past spending as a proxy for illness severity, and because less had historically been spent treating Black patients, it underestimated their need for additional services (Jain et al., 2023). Accountability is equally murky, since it's often unclear whether the developer, organization, or user is to blame when AI causes harm, though Tabassi (2023) notes proper controls can reduce that inequity.

Societal Implications **SOCIETY**

Societal impact shows up most in employment and inequality: Stanford HAI (2024) finds most people expect AI to affect their jobs, and many fear outright replacement, with AI reshaping not just which jobs exist but which skills they demand. UNESCO (2021) adds that AI can widen inequality within and between countries wherever access to it is uneven.

Sustainability Implications **SUSTAINABILITY**

Sustainability cuts both ways. Wang et al. (2024) find AI training demands significant energy, yet higher national AI development is linked to lower carbon emis-

sions, so its environmental impact needs assessing at every stage of development and use, not assumed either way.

0 GWh

ENERGY TO TRAIN A MODEL LIKE CHATGPT,
ROUGHLY 120 U.S. HOUSEHOLDS/YEAR

-0%

CARBON EMISSIONS CHANGE PER 1%
GROWTH IN AI DEVELOPMENT

Ethical Theories

Three ethical lenses help weigh these findings:

Utilitarianism

AI is ethical when it produces the greatest overall benefit for society and minimizes harm, weighing gains in medicine and efficiency against risks of discrimination and job loss.

Kantian Ethics

Human beings must be respected as ends in themselves, not merely as sources of data, whatever the efficiency gains on offer.

Social Contract Theory

Governments and organizations bear the duty to create regulations that let society share AI's benefits while minimizing potential harm.

Taken together, AI and Expert Systems can be genuinely beneficial for society when developed appropriately, but only if ethical, societal, and sustainability considerations are weighed together, not one at a time.

Figure 1: AI & Expert Systems



AI & Expert Systems

Advanced technologies that learn, reason, and support human decision-making



ETHICAL IMPLICATIONS

Ensure systems are fair, accountable, and respect fundamental rights.



Fairness & Bias

Biased training data can produce discriminatory outcomes, as seen in real health care risk algorithms.



Accountability & Transparency

Responsibility for automated decisions is often unclear; explainability is essential.



Privacy & Data Protection

Collecting and using personal data raises significant privacy and security concerns.



SOCIETAL IMPLICATIONS

AI affects people and communities in broad and unequal ways.



Impact on Employment

Potential job displacement and transformation of work requiring reskilling.



Inequality & Digital Divide

AI can widen existing social and economic inequalities when access is uneven.



Equal Access & Inclusion

Ensuring all segments of society can benefit from AI and take part in its use.



SUSTAINABILITY IMPLICATIONS

AI development and use carry real environmental consequences.



Energy Consumption

Training and operating AI systems demands substantial electricity.



Carbon Emissions

Greater energy use is linked to higher emissions and environmental strain.



Potential for Sustainability

AI can also support efficiency, climate action, and sustainable development.



RESPONSIBLE DEVELOPMENT & ADOPTION

Guided by ethics, governance, and continued human oversight.

Recommendations

1 Audit and keep a human in the loop

Regular audits should check for bias and inequity, especially where results affect people's lives, with a human kept in the loop so responsibility never rests on the automated system alone (Tabassi, 2023; World Health Organization, 2021).

2 Close the access gap

Governments and organizations should ensure every segment of society gets equal access to AI's benefits, since uneven access deepens existing inequalities (UNESCO, 2021), and prepare people for AI's growing impact on employment.

3 Develop and use AI sustainably

Sustainability should be weighed throughout development and use, cutting unnecessary energy consumption while still exploring AI's potential to support sustainability (Wang et al., 2024).

Conclusion

This paper examined the ethical, societal, and sustainability implications of AI and Expert Systems. Despite real advantages for health care, education, economics, and industry, these technologies pose serious problems: bias, privacy, accountability, job displacement, inequality, and high energy usage.

The results point to one necessity: responsible AI. Fairness, transparency, human control, and respect for human rights must guide the creation of new AI tools, alongside equal access and sustainability.

"AI and Expert Systems may positively change society greatly, if governed properly."

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0:45

References

- Jain, A., Brooks, J. R., Alford, C. C., Chang, C. S., Mueller, N. M., Umscheid, C. A., & Bierman, A. S. (2023). Awareness of racial and ethnic bias and potential solutions to address bias with use of health care algorithms. *JAMA Health Forum*, 4(6), e231197. <https://doi.org/10.1001/jamahealthforum.2023.1197>
- Stanford Institute for Human-Centered Artificial Intelligence. (2024). *AI Index report 2024*. Stanford University. <https://hai.stanford.edu/ai-index/2024-ai-index-report>
- Tabassi, E. (2023). *Artificial intelligence risk management framework (AI RMF 1.0)* (NIST AI 100-1). National Institute of Standards and Technology. <https://doi.org/10.6028/NIST.AI.100-1>
- UNESCO. (2021). *Recommendation on the ethics of artificial intelligence*. UNESCO. <https://www.unesco.org/en/legal-affairs/recommendation-ethics-artificial-intelligence>
- Wang, Q., Li, Y., & Li, R. (2024). Ecological footprints, carbon emissions, and energy transitions: The impact of artificial intelligence (AI). *Humanities and Social Sciences Communications*, 11, 1043. <https://doi.org/10.1057/s41599-024-03520-5>
- World Health Organization. (2021). *Ethics and governance of artificial intelligence for health: WHO guidance*. World Health Organization. <https://www.who.int/publications/i/item/9789240029200>

Abstract

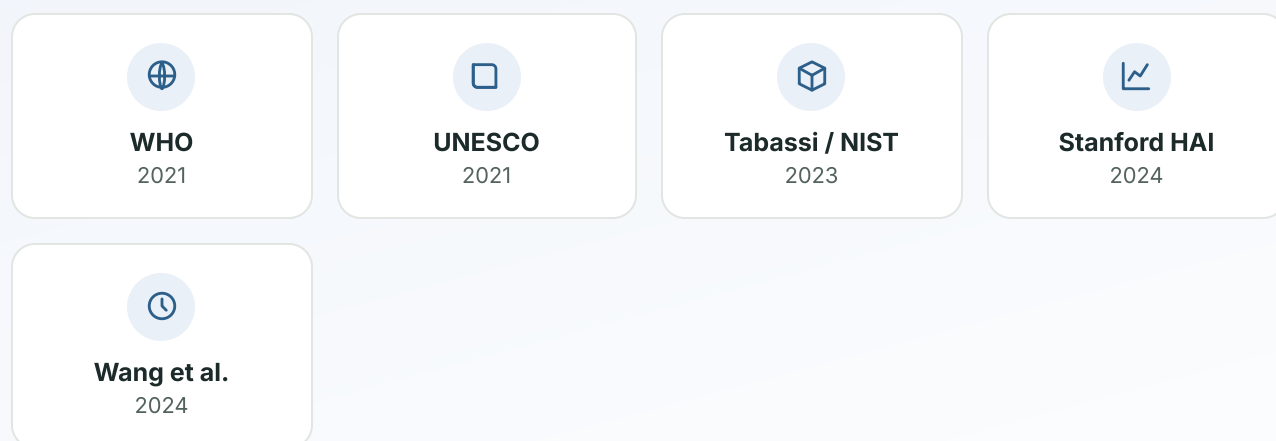
AI and Expert Systems are reshaping health care, education, finance, manufacturing, and cybersecurity through greater efficiency and better decision making. This report examines their ethical, societal, and sustainability implications, drawing on WHO, UNESCO, and NIST reports, the Stanford AI Index, and recent peer-reviewed research.

"There are ethical considerations because of issues such as skewed data, lack of accountability in AI-based decision making, and the violation of human rights."

These technologies bring real benefits, but also privacy, bias, accountability, job-loss, and inequality risks. Their sustainability impact cuts both ways too: heavy computational demand offset by AI's potential to help lower carbon emissions.

Literature Review

Current research shows AI's immense potential, especially in health care: the World Health Organization (2021) points to real progress in drug discovery, genomics, radiology, pathology, and prevention. Alongside these gains, though, researchers and international bodies increasingly flag the wider consequences of AI for individuals and society.



On ethics, the World Health Organization (2021) points to the digital divide, poor data quality, biased clinical data, and improper collection as recurring problems, alongside questions of liability when someone is harmed by an AI system. UNESCO (2021) broadens the frame further, arguing ethical AI is not just about fixing bad datasets but about protecting human dignity, well-being, and basic rights. Tabassi (2023) adds that without proper controls, AI intensifies unfair outcomes, but with them, it can help reduce them.

On employment and inequality, the Stanford Institute for Human-Centered AI (2024) finds 57% of respondents expect AI to affect their jobs within five years, and 36% fear outright replacement. UNESCO (2021) reads this as an early warning: AI can deepen inequality both within and between countries wherever access to it is uneven.

"57% of respondents believe AI would impact their jobs within five years, and 36% fear their jobs might be replaced outright."

Stanford Institute for Human-Centered AI, 2024

On sustainability, Wang et al. (2024) find the picture cuts both ways: training a model like ChatGPT consumes 1.287 gigawatt-hours of electricity, roughly the annual usage of 120 American households, yet every 1% rise in a country's AI development level is linked to a 0.0013% drop in carbon emissions. Across the literature, the throughline is the same: AI's influence on society is genuinely mixed, and responsible implementation means weighing its ethical, social, and environmental effects together, not one at a time.

Appendix

Supporting excerpts from the primary sources cited throughout this report.
Click a card to expand it.

AI Training and Energy Consumption



AI, Inequality, and Equal Access



AI Risk Management and Responsible Development



AI in Healthcare: Benefits and Ethical Challenges



AI, Race, and Complex-Needs Algorithms

